

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

Taking log of both sides, we have:

$$\begin{aligned} r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - p_t \\ &= \log\left(P_{t+1} + \frac{P_{t+1} \cdot D_{t+1}}{P_{t+1}}\right) - p_t \\ &= \log\left(P_{t+1} \left(1 + \frac{D_{t+1}}{P_{t+1}}\right)\right) - p_t \\ &= P_{t+1} + \underbrace{\log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right)}_{\equiv dy_{t+1}} - p_t \quad (1.4) \end{aligned}$$

We also have

$$\begin{aligned} dy_t &= \log\left(1 + \frac{D_t}{P_t}\right) = \log\left(1 + \frac{e^{dt}}{e^{p_t}}\right) = \log(1 + e^{dpt}) \\ \Rightarrow -dy_t &= -\log(1 + e^{dpt}) = \log(1 + e^{dpt})^{-1} \\ \Rightarrow e^{-dy_t} &= e^{\log(1 + e^{dpt})^{-1}} = \frac{1}{1 + e^{dpt}} \\ \Rightarrow \frac{1}{e^{-dy_t}} - 1 &= e^{dpt} \\ \Rightarrow \frac{1 - e^{-dy_t}}{e^{-dy_t}} &= e^{dpt} \\ \Rightarrow 1 - e^{-dy_t} &= e^{dpt} \cdot e^{-dy_t} \\ \Rightarrow \log(1 - e^{-dy_t}) &= dpt - dy_t \quad (1.5) \end{aligned}$$

Rewriting equation (1.4) and plugging (1.5), we have:

$$\begin{aligned} r_{e,t+1} - (d_{t+1} - d_t) &= (d_t - p_t) - (d_{t+1} - p_{t+1}) + dy_{t+1} \\ \Rightarrow r_{e,t+1} - \Delta d_{t+1} &= d_{p_t} - d_{p_{t+1}} + dy_{t+1} \\ &= (d_{p_t} - dy_t) - (d_{p_{t+1}} - dy_{t+1}) + dy_t \\ &= \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}}) + dy_t \quad (1.6) \end{aligned}$$

Applying first-order Taylor approximation to $\log(1 - e^{-dy_t})$ around $\bar{dy}$, we have:

$$\log(1 - e^{-dy_t}) = f(dy_t) \approx f(\bar{dy}) + f'(\bar{dy})(dy_t - \bar{dy})$$

Define $K \equiv e^{-\bar{dy}}$. We have the following:

$$f(\bar{dy}) = \log(1 - e^{-\bar{dy}}) = \log(1 - K)$$

$$f'(\bar{dy}) = \frac{1}{1 - e^{-\bar{dy}}} \cdot e^{-\bar{dy}} = \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}} = \frac{K}{1 - K}$$

As a result, $f(dy_t) = \log(1 - K) + \frac{K}{1 - K}(dy_t - \bar{dy})$

Similarly, $f(dy_{t+1}) = \log(1 - K) + \frac{K}{1 - K}(dy_{t+1} - \bar{dy})$

Plugging these equations into (1.6), we have:

$$\Rightarrow r_{e,t+1} - \Delta d_{t+1} = \log(1 - K) + \frac{K}{1 - K}(dy_t - \bar{dy}) - \log(1 - K) - \frac{K}{1 - K}(dy_{t+1} - \bar{dy}) + dy_t$$

$$= \frac{K}{1 - K}(dy_t - dy_{t+1}) + dy_t$$

$$= \frac{K + 1 - K}{1 - K} dy_t - \frac{K}{1 - K} dy_{t+1}$$

$$= \frac{1}{1 - K} dy_t - \frac{K}{1 - K} dy_{t+1}$$

$$= \frac{dy_t - K dy_{t+1}}{1 - K}$$

$$\Rightarrow dy_t = (1 - K)(r_{e,t+1} - \Delta d_{t+1}) + K dy_{t+1}$$

⇓ Recursively substituting dy_{t+h}

$$dy_t = (1 - K) \left[\sum_{h=1}^H K^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H K^{h-1} \cdot \Delta d_{t+h} \right] + K^H \cdot dy_{t+H} \quad (1.7)$$

Taking conditional expectation of both sides and assuming that $E_t[X] = X$, we have:

$$dy_t = (1 - K) \left[\sum_{h=1}^H K^{h-1} \cdot E_t[r_{e,t+h}] - \sum_{h=1}^H K^{h-1} \cdot E_t[\Delta d_{t+h}] \right] + K^H \cdot dy_{t+H} \quad (1.8)$$

knowing $0 < k < 1$ by construction & assuming the no-bubble condition holds such that $\lim_{H \rightarrow \infty} k^H \cdot dy_{t+H} = 0$,

we let $H \rightarrow \infty$ in equation (1.8) and have:

$$dy_t = (1-k) \left[\sum_{h=1}^{\infty} k^{h-1} \cdot E_t[r_{e,t+h}] - \sum_{h=1}^{\infty} k^{h-1} \cdot E_t[\Delta d_{t+h}] \right] \quad (1.9)$$